AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM275
Parallel and Distributed Systems

Final Examination
Spring Semester 2025/2026

Date: 06/Jun/2026

Number of Exam Pages: 2 (including this cover sheet)	**Time Allowed: (2) Hours**

Instructions:

- Total Marks: **50**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

AOU EGYPT

Part I: Multiple Choice Questions (10 Marks, 2 marks for each)

1. Which theorem states that it is not possible to guarantee all three of Consistency, Availability, and Partition Tolerance simultaneously?
a) Moore's Law
b) CAP Theorem
c) Amdahl's Law
d) Flynn's Taxonomy

2. What is the default port used for Cassandra client connections?
a) 8080
b) 9042
c) 27017
d) 3306

3. Which authenticator stores encrypted credentials in system tables and enables username/password authentication?
a) AllowAllAuthenticator
b) PasswordAuthenticator
c) LDAPAuthenticator
d) SimpleAuthenticator

4. Which cloud service model provides the most control over IT resources?
a) SaaS (Software as a Service)
b) PaaS (Platform as a Service)
c) IaaS (Infrastructure as a Service)
d) FaaS (Function as a Service)

5. What is an Amazon Machine Image (AMI)?
a) A virtual machine running in the cloud
b) A template used to create an EC2 instance, containing an OS and sometimes pre-installed software
c) A storage volume
d) A security group

Part II: Short Answer Questions (15 marks = 10+5)

1. Describe the roles of partition key and clustering key in Cassandra. How do they affect data distribution and query performance? Provide an example.
2. What is AWS and what are the three primary ways to interact with AWS services? Describe each method and when you might use it.

Part III: Problem-Solving Question (25 marks = 15+5+5)

1) Write CQL commands to insert the following student records into the students table:

student_id	first_name	last_name	email	enrollment_date	gpa
230122	John	Doe	john.doe@email.com	2023-09-01	3.75
230157	Jane	Smith	jane.smith@email.com	2023-09-01	3.90
220365	Bob	Johnson	bob.j@email.com	2022-01-15	3.25

2) Verify the inserts
3) Update Bob Johnson's GPA to 3.50

End of questions

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